

D3CODE 2025 – India Edition Idea:

Quick Docs – “Personal AI Voice Assistant for Files”

Problem: In today’s fast-paced world, accessing and extracting information from documents is time-consuming and inefficient. People often struggle with:

- Searching for specific answers inside large PDFs or documents.
- Reading entire documents when they only need a small piece of information.
- Managing document overload – from academic papers, legal files, reports, notes, etc.
- Collaboration and sharing information quickly and securely.
- Information retrieval during emergencies where time is critical.

This results in lost productivity, delays, and frustration across sectors like education, healthcare, business, and governance.

Our Solution:

Quick Docs is a **mobile application powered by AI** that allows users to **upload, search, view, and extract answers** from documents instantly. Users can store files safely, access them on the go, and an AI-Driven **Voice-to-Voice File Assistant**, supporting **Regional Language Responses** like **Telugu, English, Hindi** and Chatbot replies based on user file data effectively using Retrieval Augmented Generation and Optical Character Recognition (OCR).

Objective:

The objective of **Quick Docs** is to develop an **AI-powered mobile application** that serves as a **trusted document companion** for people around the world, especially those who want instant information from tons of important files. Quick Docs provides **personalized, localized, and culturally relevant** information using advanced Large Language Models (LLM) enabling people to get accurate data without relying on manual searches.

Implementation (Technologies Used):

- **React Native + Expo** – Cross-platform mobile app
- **LLMs (OpenAI/Gemini)** – For real-time document Q&A
- **Google Vision API** – For OCR and Text Extraction.
- **Firebase & MongoDB** – Authentication, Notifications and Database

- **Text-to-Speech, Speech-to-Text & Voice Recognition** – For Regional Language response ability & accessibility
 - **Multilingual NLP Support** – English, Telugu, Hindi (to start)
 - **AES Encryption** – Enabling Secure and scalable CRUD APIs with encrypted data.
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Key Screens/ Modules:

1. **Home Screen** – View, Download, Share functions of stored files of user.
 2. **Upload Screen** – Uploading supported format files (.pdf, .docx, .jpg, .jpeg, .png).
 3. **AI Chatbot Screen**– Retrieves and answers text-based responses through chat.
 4. **Voice Assistant Screen** – Personalised Voice to Voice Regional Language Supported Assistant (launched with beta languages like Telugu, English, Hindi).
 5. **Premium Screens** – Priority is allocated based on 3 variant Premium Subscriptions.
 6. **Data Analytics Screen** – Visualising the recent searches, number of prompts used.
 7. **Profile Screen** – User details, special components for premium users.
 8. **Authentication Screens** – Login, Signup, Forgot Password Screens are validated.
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Workflow:

1. Uploading Documents

- The user can upload files in multiple formats including:
 - PDF
 - JPG / JPEG / PNG
 - DOCX
 - Once uploaded, the document undergoes:
 - **Parsing & Content Extraction** using:
 - **Optical Character Recognition (OCR)**
 - **Google Cloud Vision API**
 - The original file and processed data are stored in **Google Cloud Storage** in a secure and encrypted format.
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2. Secure Storage & Data Encryption

- Document content and metadata are encrypted using:

- **AES encryption standard**
 - AES keys and algorithms for secure data handling.
 - Sensitive data stored in **MongoDB** is fully encrypted, ensuring privacy and security.
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3. Document Access & Management

- Users can view uploaded files in:
 1. **Home Screen**
 2. **View File Screen**
 - Available actions for each file:
 - **View** – Open and read the document.
 - **Download** – Save the file for offline access.
 - **Delete** – Remove the file from the cloud.
 - **Share** – Send the file to other applications or users.
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4. AI Chatbot – Instant Information Retrieval

- The user can access the **AI Chatbot Screen** to:
 - Ask questions in text format.
 - Get instant answers by searching through vector embeddings of uploaded files (all supported types) and indexed data efficiently.
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5. Voice-to-Voice Assistant

- The **Voice Ask Assistant Screen** enables interaction with AI through voice commands.
- Key features:
 - Supports communication in regional languages:
 - Telugu (Beta)
 - Hindi (Beta)
 - English (Beta)
 - The AI responds in the same language as the query.
- This feature is available only under the **Highest Priority Premium Subscription**.

6. Profile Management

- The **Profile Screen** allows users to:
 - View personal data.
 - Update information.
 - Manage subscription and settings.
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7. Premium Features

- Three premium subscription levels, with differentiated access.
 - The **Voice-to-Voice Assistant** is unlocked only with the **Highest Priority Subscription**.
 - Premium access provides:
 - Enhanced services.
 - Priority support.
 - Expanded AI functionalities.
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Hackathon Deliverable:

- Working app prototype of **Quick Docs** with:
 - Real-time Personalised AI chatbot (User specific).
 - Automatic Detection of user's voice and responding in same language.
 - Voice assistant in Telugu, English, Hindi (Beta Version)
 - Premium Subscription as a Commercial aspect.
 - Push notifications for important events.
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End Users by Category & How Quick Docs Helps Them:

1. Business People

- **Use Case:**
 - Quickly access reports, invoices, and market research documents.
 - Ask for summaries or specific data without navigating complex files.
- **How It Helps:**
 - Saves time in searching for key financial or operational information.
 - Allows seamless communication in preferred languages during meetings.

2. Students

- **Use Case:**
 - Study from textbooks, lecture notes, and PDFs without reading entire documents.
 - Ask for definitions, formulas, or explanations in their native language.
 - **How It Helps:**
 - Reduces cognitive load and improves learning efficiency.
 - Makes education more accessible, especially in regional areas with language barriers.
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3. Working Professionals

- **Use Case:**
 - Manage multiple documents like project files, manuals, and reports.
 - Ask for instructions, procedures, or data points while commuting or multitasking.
 - **How It Helps:**
 - Enhances productivity by providing instant, accurate information.
 - Reduces dependency on manual search or human support.
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4. Teachers

- **Use Case:**
 - Access teaching material quickly, prepare lessons, or explain concepts.
 - Communicate with students in their regional language during virtual classes.
 - **How It Helps:**
 - Facilitates efficient lesson planning and content delivery.
 - Encourages better student engagement, especially in rural or multilingual settings.
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5. Healthcare Workers

- **Use Case:**
 - Retrieve patient histories, medical reports, or treatment guidelines instantly.

- Communicate medical instructions with patients in their own language during emergencies.
 - **How It Helps:**
 - Saves crucial time in healthcare decision-making.
 - Bridges communication gaps in multilingual environments.
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6. Agriculture & Field Workers

- **Use Case:**
 - Access agricultural/ weather reports/ market prices via voice queries.
 - Get real-time advice in their regional language while working in the field.
 - **How It Helps:**
 - Empowers farmers and field workers with accessible information.
 - Enhances productivity, decision-making without needing technical literacy.
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Post-Campaign Development

After the hackathon, we will:

1. **Expand regional support** – Add Tamil, Marathi, Bengali, Kannada, etc.
 2. **Add Vision-AI Features** – Custom OCR application for cost effectiveness.
 3. **Custom LLM** – Moderated & Custom LLM deployment in our own server.
 4. **Offline-first capability** – For low-network regions with sync feature
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Final Impact

Quick Docs will become India's **first AI-powered, personalised assistant** that supports multilingual voice-to-voice AI assistant. It is a game-changer for various sectors. By enabling seamless, natural communication in multiple languages, it breaks down barriers of literacy, access, and efficiency. Whether it's business, education, healthcare, or agriculture, this feature empowers users with instant, accurate, and secure information retrieval—right from their fingertips.

Submitted by,

Team: **Unstoppable Force**

Rajeswar Yadav | Jyothi Swaroopa | Lova Reddy